Drug Administration

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Objective



By the end of lecture, the student will be able to:

1. Identify a brief introduction about drug and medication .
2. List the essential principles for the administration of medications.
3. Discuss the steps of the different routes of administration of medications.
4. Calculate a safe drug dosage needed for the individual child.

Introduction:

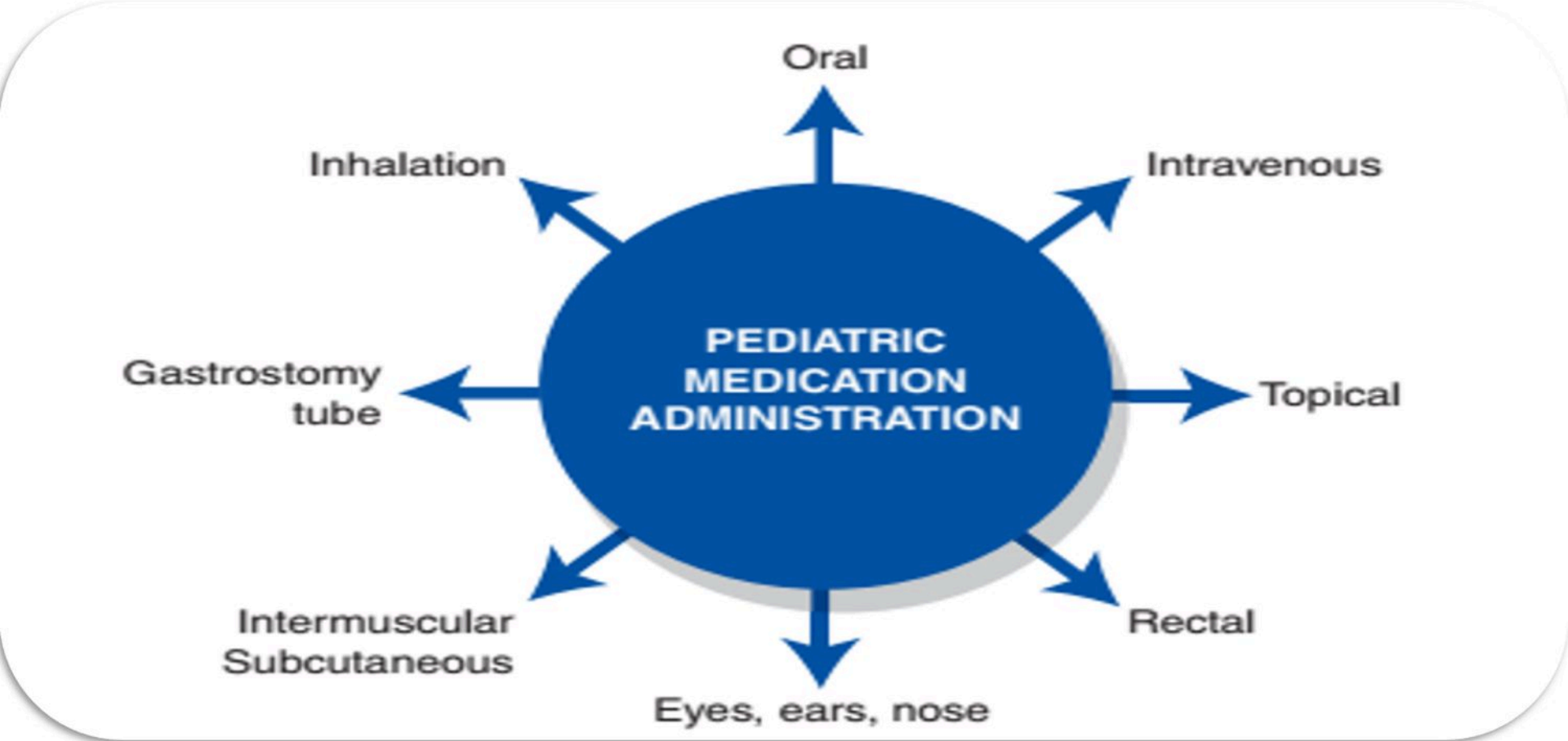


- child responds to drugs more rapidly and unpredictably than an adult does for this reason; **pediatric drug administration requires special care**. Such factors as **age**, **weight**, **body surface area**, and **drug form** and **route** may dramatically affect a child's response to a drug. For example, because of his thin epithelium, a neonate or an infant absorbs topical medications much faster than an older child does.

General steps for administer medication

- Assess the child's condition to determine the need for the medication and the effectiveness of previous therapy.
- Carefully observe the child for a rash, pruritus, cough, or other signs of an adverse reaction to a previously administered drug.
- Identify the child by comparing the name on his wristband with that on the medication card. If the child can talk and respond, ask him his name.
- Explain the procedure to the child and his parents. Use terms the child can understand.
- Provide privacy, especially for an older child.

Drug Administration Routes



Six right for administer medication

- 1-Right Medication
- 2-Right Patient
- 3-Right Dosage
- 4- Right Route
- 5- Right Time
- 6-Documentation

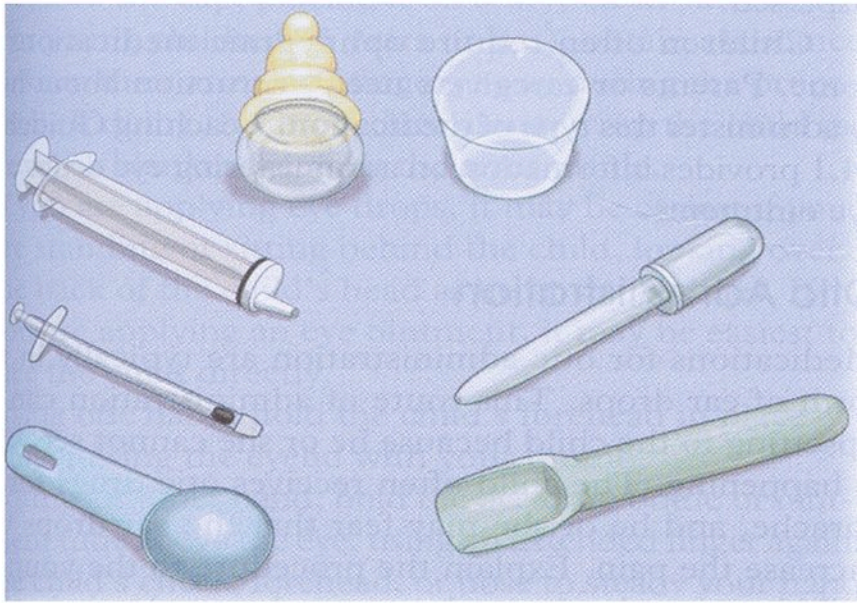
Administering Oral Medications



The syringe or cup method of administering oral medications to infants or children. Note that one of the infant's arms is behind the nurse's back while the nurse restrains the other. The legs of the infant can be held firmly between the knees of the nurse for better control during medication administration.

Equipment for oral medications

1. Prescribed medication
2. Plastic disposable syringe, plastic medicine dropper, or spoon
3. Medication cup
4. Water, syrup, or jelly (for tablets)
5. Optional: fruit juice.



Steps of Oral Administration



1. Verify medication orders and assemble supplies.
2. Place the infant's legs between your knees to maintain control of them.
3. Place one of the infant's arms behind your back and restrain the other arm by "**hugging" the child** (this helps to prevent the child from tipping the medication cup).
4. Give the medication slowly by cup or syringe to allow the child time to swallow. An older child may be given the choice to self-administer the medication by cup or syringe.

Cont...

5. Provide “chasers” of water, fruit juice, or frozen ice-pops to help lessen the residual taste of the medicine. The patient’s age and diet prescription are considered when choosing a chaser.

6. Document in the medication administration record (MAR) the medication, route, time, and dose given

- **Measure a drug ordered in teaspoons in milliliters; the established standard is 5 ml/tsp.**



Notices

- Medication is administered slowly, especially if the child is crying. The child's head and shoulders are elevated to prevent aspiration. Toddlers may attempt to push away the medicine cup. In anticipation of this response, the nurse holds the child in a "**hug" position in the nurse's lap** in a semi sitting position
- The use of important sources of nutrients, foods, or liquids (e.g., orange juice) for disguising the taste of the medication is discouraged because the child may develop a distaste for these foods.
- If a nasogastric tube is in place, the nurse tests for proper placement of the tube before pouring medication into the funnel. A small amount of water is administered afterward to flush (cleanse) the tube.

- Never pinch the infant or child’s nostril when administrating medications
- Don’t place medication in a baby’s bottle
- Suspension must be administered immediately after measurement to prevent settling and giving an incomplete dose
- Slide the syringe or medicine dropper into the infant's mouth alongside his tongue. Release the medication slowly
- If the medication is in tablet form, first crush the tablet, if appropriate, and mix it with water, syrup, or jelly. Use a spoon, syringe, or dropper to administer the medication. (Toddler)
- Giving medication in tablet or capsule form for older enough child (between ages 4 and 6), you should teach him how to swallow solid medication.

Administering parenteral medications

- **Nose Drops**
- **Ear Drops**
- **Eye Drops**

the principles for administering nose drops, ear drops, and eye drops to children are essentially the same as for adults. Infants and small children may need to be restrained in a mummy restraint

NURSING Tips

- **To reduce unpleasant sensations when administering medications:**
- **Eye**—Apply finger pressure to the **lacrimal punctum at the inner aspect of the eyelid** for 1 minute to prevent drainage of medication to the nasopharynx and the unpleasant "tasting" of the drug.
- **Ear**—Allow medications stored in the refrigerator to **warm to room temperature** before instillation.
- **Nose**—Position the child with the **head hyperextended** to prevent strangling sensations caused by medication trickling into the throat rather than up into the nasal passages.

Nose drops



Tilt the head down and back to instill nose drops.

Steps

1. Immobilize the infant with a mummy restraint.

2. Wipe excess mucus from nose with a tissue.

3. Place the infant on his or her back, with the head over the side of the mattress or the neck extended over a pillow.

4. Encircle the infant's cheeks and chin with the left arm and hand to steady.

5. Instill drops with the right hand.

6. Keep infant in this position for 30 seconds to 1 minute to allow the drops to reach the proper area.

7. Remove restraints. Make the infant comfortable.

8. Remove gloves and perform hand hygiene

9. Record: time, name of nose drops, strength, and number of drops instilled, how the patient tolerated the procedure, and untoward reactions.

Ear drops

A. In children **less than 3 years** of age, pull the pinna (earlobe) of the affected ear down and back to straighten the canal.



B. In **older children**, pull the upper pinna (auricle) up and back to straighten the canal.



Steps

1. Place child in supine position with unaffected ear down.

2. Instill the ordered number of drops

3. Gently massage the area in front of the ear to facilitate entry of the drops.

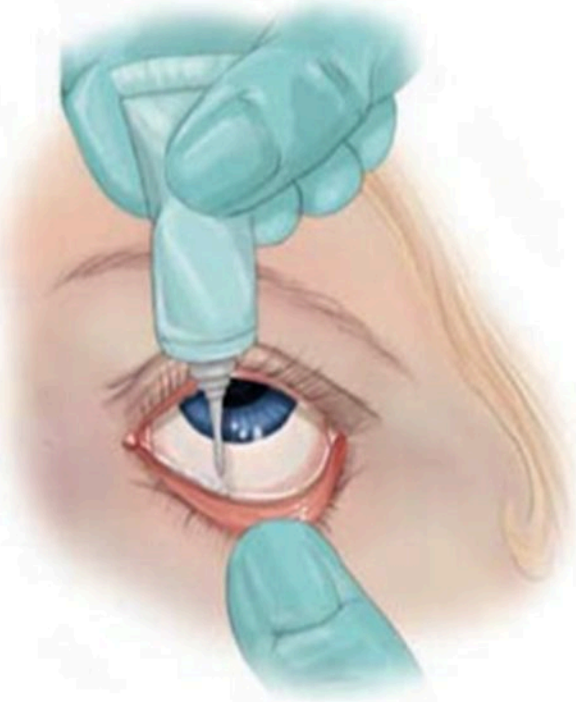
4. Keep the patient in a supine position for a few minutes to permit the fluid to be absorbed.

5. Remove gloves and perform hand hygiene.

6. Document the time, name of drug, number of drops administered, area (right or left ear), untoward reactions, and whether or not the patient obtained relief.

Eye drops and creams





Steps:



- Ophthalmic medication is administered to a child in the same manner as for the adult
- The child is informed of the need for the medication.
- The patient is identified, and the orders and the label on the bottle are checked for correct medication and concentration.
- The nurse ascertains which eye requires treatment.
- Hand hygiene is performed before and after the procedure, and clean gloves are worn. With the thumb and index finger, gentle pressure is applied in opposite directions to open the eye.
- The older child is instructed to "look up." The nurse supports his or her hand on the patient's forehead, and instills the medication into the center of the lower lid (conjunctival sac)
- The child is instructed to close the eye but not to squeeze it, because this could expel some of the solution.

Ointment

- Ointment is applied to the **same conjunctival sac** as eye drops.
- The mummy restraint is applied when only one nurse is available and the patient is an infant.
- Occasionally children refuse to open their eyes.
- The nurse must use ingenuity to coax reluctant children. It may help to involve the parents.



Inhaled Medication



- **Equipment**
- A pressurized canister
- A plastic holder
- A cap.
- Often a spacer or special tube that attaches to the inhaler is used to increase the amount of medication delivered to the lungs per spray. A spacer has either **a mouthpiece or a mask** for ease of administration and is specific to the child's developmental need.

Inhaled Medication



3-7 Child using metered-dose inhaler with spacer and face mask.





Administering inhaled medication to children through a nebulizer has the advantage of delivering a medication directly to the needed site of action.

Procedure (steps)

- Placing the mouthpiece fully into the mouth between the teeth
- Hold the inhaler in the upright position and close the lips around the mouthpiece
- Press down on the canister to dispense the medication into the spacer
- Instruct the patient to breathe deeply for 3–5 seconds through the mouth, and then hold the breath for 10 seconds.
- Remove the spacer from the mouth and breathe out slowly.
- Patient should rinse the mouth out with water after each use of inhaled steroids to help prevent oral infections, and the spacer should routinely be washed with warm soapy water and air dried
- Teach families to consistently use proper technique with their children
- Often young children resist the mask when it is first introduced; however, with consistent practice eventually they will cooperate when they realize the parent is serious about giving their medications effectively.

Administering Intramuscular Injections

- **Equipment:**

✓ Prescribed medication

✓ Appropriately sized syringe and needle

✓ Alcohol pads or povidone-iodine solution

✓ Gloves

✓ Gauze pads

✓ Cold compresses

✓ Adhesive bandage.



TABLE 19-2 Age-Based Guidelines for Selecting Needle Length and Gauge

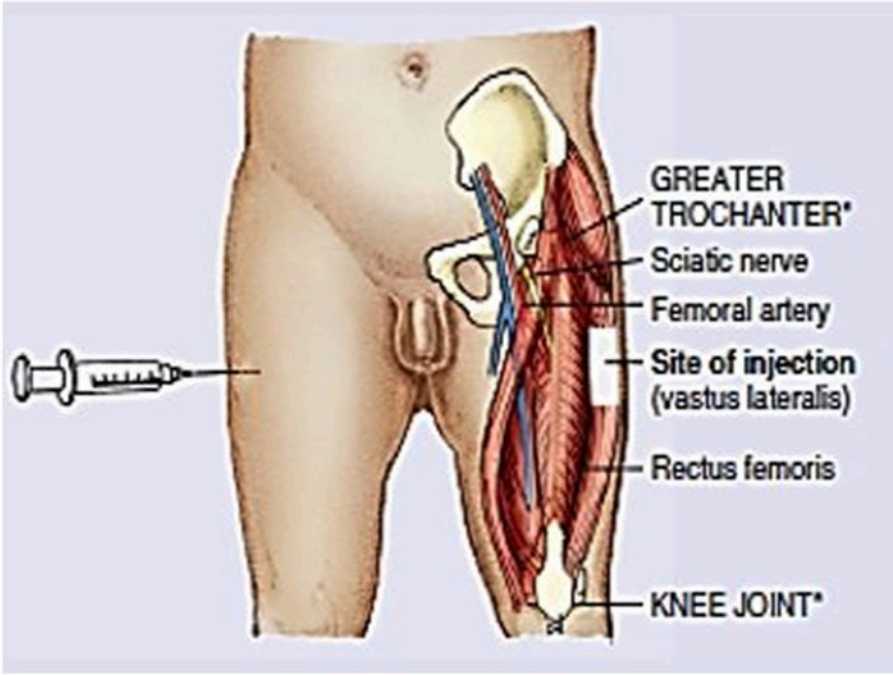
AGE GROUP	NEEDLE LENGTH	NEEDLE GAUGE
Infant	5/8 inch	25–27
Toddler or Preschool	1 inch	22–23
School-age or adolescent	1–1½ inch	22–23

I.M. INJECTION SITES IN CHILDREN

- When selecting the best site for a child's I.M. injection, consider:
  - ➢ the child's age, weight, and muscular development
  - ➢ the amount of subcutaneous fat over the injection site
  - ➢ the type of drug you're administering
- the drug's absorption rate

Vastus lateralis and rectus femoris

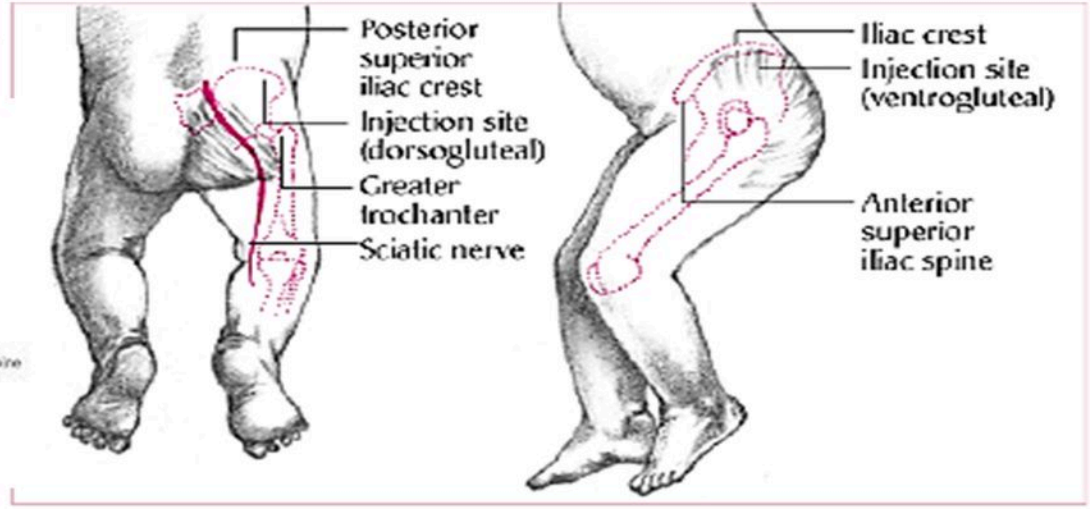
- For a child under age 3, you'll typically use the vastus lateralis or rectus femoris muscle for an I.M. injection. Constituting the largest muscle mass in this age-group and have fewer major blood vessels and nerves.



Ventrogluteal and dorsogluteal

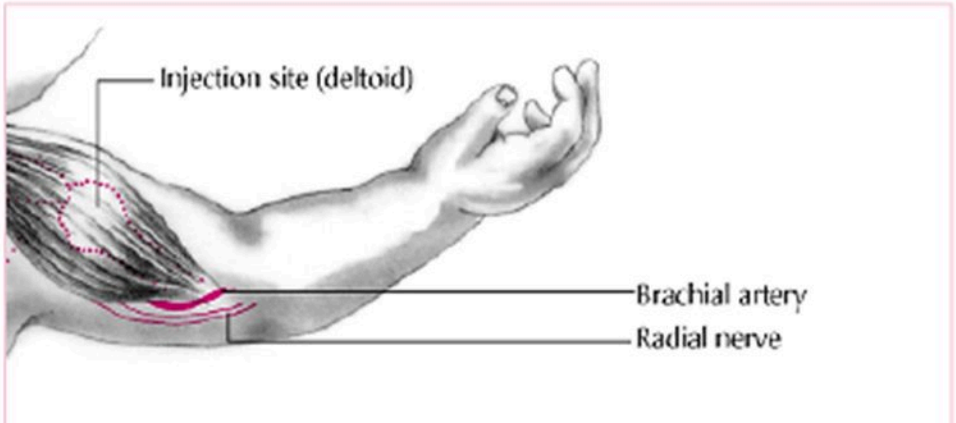
- For a child who can walk and is over age 3, use the ventrogluteal and dorsogluteal muscles. Like the vastus lateralis, the ventrogluteal site is relatively free from major blood vessels and nerves. Before you select either site, make sure that the child has been walking for at least 1 year to ensure sufficient muscle development.

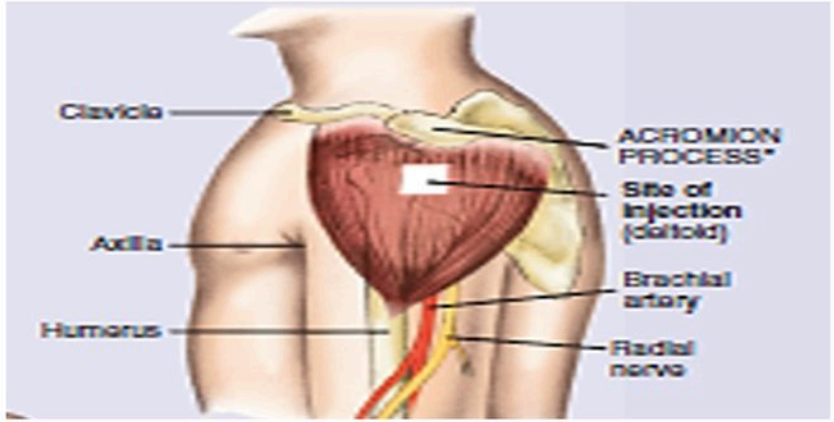




Deltoid muscle

- For a child older than 18 months who needs rapid medication results, consider using the deltoid muscle for the injection. Because blood flows faster in the deltoid muscle than in other muscles, drug absorption should be faster. However, be careful when using this site because the deltoid doesn't develop fully until adolescence. In a younger child, it's small and close to the radial nerve, which may be injured during needle insertion.





Procedure steps

- Clean the injection site with an alcohol or povidone-iodine pad. Wipe outward from the center with a spiral motion to avoid contaminating the clean area.
- Prepare medication for injection. Use a 1mL syringe and in, 25gauge needle to give the injection.
- Put on gloves. **R:**Gloves protect the nurse from contamination by blood.
- Choose an injection site that's appropriate for the child's age and muscle mass. The best site for newborn intramuscular injections is the vastus lateralis muscle.
- Stabilize the leg firmly while grasping the thigh between the thumb and fingers. To prevents sudden movement by the infant and possible injury



FIG 22-13 Holding a small child for intramuscular injection. Note how the nurse isolates and stabilizes the muscle.



- Insert the needle at a 90—degree angle(IM). Inject the medication.
- Withdraw the needle and apply gentle pressure to the site with small gauze pad to stimulate circulation and enhance absorption
- Insert the needle quickly, using a darting motion. If you're using the ventrogluteal site, insert the needle at a 45-degree angle toward the knee.
- Aspirate the plunger to ensure that the needle isn't in a blood vessel. If no blood appears, inject the medication

Giving a subcutaneous injection (SC)

- Put on gloves, and prepare the injection site with alcohol or povidone-iodine solution according to the patient's needs and hospital policy.
- sites: the middle third of the upper outer arm, the middle third of the upper outer thigh, or the abdomen. You may apply a cold compress to the injection site to minimize pain.
- Pinch the tissue surrounding the site between your index finger and thumb to ensure injection into the subcutaneous tissue.
- Holding the needle at a **45- to 90-degree angle**, quickly insert it into the tissue.
- Release your grasp on the tissue, and slowly inject the medication.
- Remove the needle quickly to decrease discomfort.
- Unless contraindicated, gently massage the area to facilitate the drug's absorption.

Giving an intradermal injection

- ✓ Put on gloves and pull the patient's skin taut (the site of choice is the inner aspect of the forearm).
- ✓ Insert the needle, bevel up**, at a 10- to 15-degree** angle just beneath the outer skin layer.
- ✓ Slowly inject the medication, and watch for a bleb to appear. Quickly remove the needle, being careful to maintain the injection angle.
- ✓ if the injection is related to allergy testing—draw a circle around the bleb, and avoid massaging the area to avoid interfering with test results.

I.V. Therapy

- **Introduction**
- In children, I.V. therapy may be prescribed to administer medications or to correct a fluid deficit, improve serum electrolyte balance, or provide nourishment. Primary nursing concerns related to pediatric I.V. therapy include correlating the I.V. site and equipment with the reason for therapy and the patient's age, size, and activity level. For example, a foot or scalp vein I.V. site may be used for infants, whereas a peripheral hand or wrist vein may be more suitable for ambulatory older children.
- During I.V. therapy, the nurse must continually assess the patient and the infusion to prevent fluid overload and other complications

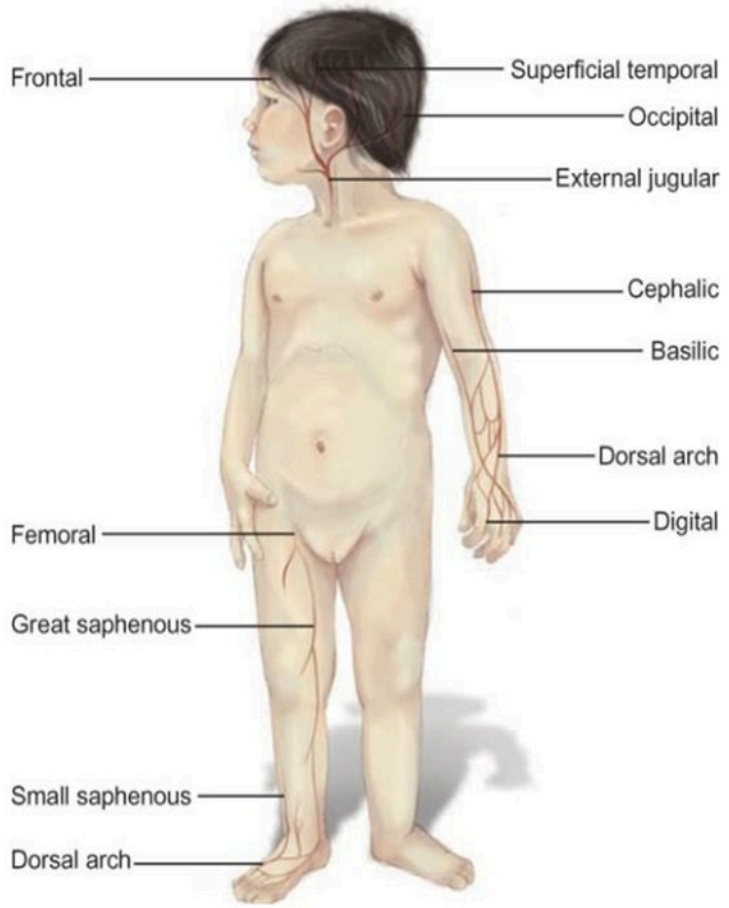
Equipment



- ❖ Prescribed I.V. fluid
- ❖ volume-control set with micro drip tubing
- ❖ infusion pump
- ❖ I.V. pole
- ❖ normal saline solution or sterile dextrose 5% in water (D5W) for injection
- ❖ povidone-iodine solution
- ❖ alcohol pads
- ❖ 3-ml syringe
- ❖ child-size butterfly needle or I.V. catheter
- ❖ tourniquet • ½″ or 1″
- ❖ sterile tape gloves

Common pediatric I.V. sites

- Below are the most common sites for I.V. therapy in infants and children. Peripheral hand, wrist, or foot veins are typically used with older children, whereas scalp veins are used with an infant.



Procedure

- Match the name on the patient's wristband with the name on the physician's order (or on the medication card). Ask the patient (or his parents) whether he's allergic to povidone-iodine solution or to any type of tape.
- Explain the reason for the I.V. therapy.
- Be sure to have a staff member available to assist you.
- Wash your hands and put on gloves.
- Select the insertion site for the butterfly needle or catheter, and avoid placing the I.V. line in the patient's dominant arm or in areas of flexion if possible. Avoid previously used or sclerotic veins.
- To locate an appropriate scalp vein, carefully palpate the site for arterial pulsations. If you feel these pulsations, select another site.

- To find an appropriate peripheral site, apply a tourniquet to the patient's arm or leg and palpate a suitable vein.
- If you're inserting a butterfly needle, flush the tubing connected to the butterfly with D5W or normal saline solution.
- Clean the insertion site. Unless contraindicated, use a pad containing povidone-iodine solution. Wipe with a circular motion from the insertion site's center to the outer rim. Let the solution dry.
- Insert the I.V. needle into the vein. Watch for blood to flow backward through the catheter or butterfly tubing, which confirms that the needle is in the vein.
- Loosen the tourniquet, and attach the I.V. tubing to the hub of the needle or catheter. Begin the infusion.
- Secure and protect the I.V. line as needed.

Complications of I.V therapy

1. Infection
2. Fluid Overload
3. Electrolyte Imbalance
4. Infiltration
5. Circulatory Impairment

Administration of safe medication:

- To administer medications safely, the following actions are necessary:

1. Accurately interpret the doctor's order.
2. Accurately calculate the amount of drug to be given for the prescribed dose.
3. Using the five rights for drug administration such as:

- Right medication, Right dose, Right patient, Right time and Right route of administration.

1. Document medication administration according to agency policy.

Calculating Safe Dosages for Children

- **STEPS**
- Dosage based on weight in kilograms

- Its necessary, convert child's weight in pounds to kilograms. Child weighs 22 pounds.
  - Conversion: 1 kilogram (kg) = 2.2 lb
  - 22 lb/2.2 = 10 kg

- Review medication reference to determine recommended range in
- milligrams/kilogram (mg/kg)

Drug Calculation

- To calculate a fractional drug dose use the formula:

$$\frac{Dose\ on\ hand}{Quantity\ on\ hand} = \frac{Desire\ dose}{Quantity\ Desire}$$

(X= $\frac{Desire\ dose}{Dose\ on\ hand} * Quantity\ on\ hand$)

- For example, you have a prescription for 90 mg acetaminophen. Acetaminophen is supplied as 125 mg drug in 5 ml liquid. Using the formula:
- $\frac{125\ mg}{5\ ml} = \frac{90\ mg}{X}$
- Cross multiply: X= $\frac{90\ mg * 5\ ml}{125\ mg}$ =3.6 ml

$$\frac{90\ mg}{125\ mg} \times 5\ ml = 3.6\ ml$$

•Reference text states recommended dosage range is 30 to 50 mg/kg/day in 4 divided doses.
•Multiply child's weight by the lower and upper limits of the dosage range.
• 10 kg × 30 = 300 mg/day
• 10 kg × 50 = 500 mg/day
•Child's recommended range = 300 to 500 mg/day
•Compare physician's order with dosage range to determine whether medication dosage is safe.
•Order reads: Give 500 mg q6h.
- 500 mg given q6h = 500 × 4 = 2000 mg/day
•Conclusion: Dosage exceeds recommended range.

- Dosages outside recommended range should be verified with physician. "Dr. Ahmed, I've just reviewed the new medication order for Sara. According to my calculations, the dosage exceeds that recommended for his weight according to both the formulary and pharmacist.

Dosage Calculation (Home work)

- Order: Amoxil (amoxicillin) 70 mg PO q8h.
- Patient: Child weighs 22 lbs.
- Safe dose is 20–40 mg/kg/day in divided doses q8h.
- Available as: 125 mg/5 mL
- A. Is the dose safe? _____
- B. If safe, how much liquid will you give the child per dose? _____
- C. If not safe, what is the range of safe doses for this child? _____







Thank you for listening

